Frozen-backbone JEPA-style probes on HAM10000: preliminary results from a nine-experiment ablation, with an unpartitioned contamination caveat

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github.com/AbdelStark/derma-jepa

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Abstract

Self-supervised representations promise robustness to nuisance variation; whether that robustness transfers across novel nuisance families is rarely tested. We probe this on a leakage-controlled HAM10000 longitudinal proxy with three disjoint synthetic nuisance families — partitioned along the deterministic- augmentation axis — training a JEPA-style latent predictor on stable pairs from two of three families and evaluating on the third unseen family. We report a preliminary nine-experiment arc. Across four frozen-natural-image configurations spanning two pretraining paradigms within the ViT-B class (DINOv2 ViT-B/14, OpenAI CLIP ViT-B/16), two predictor scaffolds (linear with identity warm-start, 2-layer identity-residual MLP), and two optimisers (SGD, Adam), test AUROC on the held-out family lands in 0.25–0.29 — below random and below the strongest oriented baseline (pixel L2 at 0.580). A frozen general-medical backbone (BiomedCLIP, PMC-15M scientific-article figures, with no documented raw HAM10000/ISIC archive ingestion) lifts test AUROC by only +0.04 to 0.329 ± 0.012 across 5 seeds. A frozen dermoscopy-pretrained backbone (DermLIP, Derm1M) lifts test AUROC to $\mathbf{0.944 \pm 0.003}$ across 5 seeds, +0.363 above the strongest baseline and the only above-random result on the held-out family across nine runs. Holding architecture (ViT-B/16) and predictor (linear) constant, varying only the pretraining-data domain produces a +0.66 end-to-end AUROC swing (web $\rightarrow$ dermoscopy) that is concentrated in the dermoscopy-specific step (+0.62 of +0.66, with the residual +0.04 at the web $\rightarrow$ general-medical step). **We explicitly do not claim that this swing is dermoscopy-domain transfer**: DermLIP's Derm1M corpus almost certainly contains HAM10000, and the experiments reported here cannot separate dermoscopy-domain transfer from HAM10000 image-level overlap. We release the methodology, the configurations, and the run archive, and sketch a candidate partition experiment (a non-HAM10000 dermoscopy SSL pretrain) as a natural next step we may consider after gathering community feedback on the methodology and the preliminary results. The contribution is the proxy-task design, the failure characterisation, and a clearly scoped open question — not a conclusion about whether frozen-backbone JEPA generalises on dermoscopy.

1 Introduction

Lesion monitoring is fundamentally a change-detection problem under heavy nuisance variation. Repeated photographs of the same skin lesion vary because of illumination, framing, hair occlusion, camera optics, and compression; clinically meaningful change is small relative to that nuisance envelope. Self-supervised learning (SSL) representations are routinely pitched as broadly transferable across image distributions and tasks (Caron et al., 2021; Oquab et al., 2024; Radford et al., 2021) — a property we read, perhaps charitably, as approximate invariance to many of the photometric and framing nuisances we care about here. The joint-embedding predictive architecture (JEPA) family (LeCun, 2022; Assran et al., 2023) explicitly framing prediction in representation space as the path to invariance. Our question is narrow: *when a JEPA-style latent predictor over a frozen vision backbone is trained on stable-pair invariance with respect to one set of synthetic nuisance operations, does that invariance generalise to a disjoint nuisance family the predictor never saw at training time?*

We probe this on HAM10000 (Tschandl et al., 2018). HAM10000 is cross-sectional, so we construct a leakage-controlled longitudinal proxy: lesion-ID-aware splits, synthetic stable-pair augmentations applied post-split, and changing-pair construction matched on diagnosis and anatomical site. Three deterministic synthetic nuisance families — `strong`, `strong_held_out`, and `strong_held_out_2` — are designed to share no operation type with each other. We train on stable pairs drawn from {`strong`, `strong_held_out`} and evaluate on a third unseen family `strong_held_out_2`. The predictor is a small function $g_\theta : \mathbb{R}^d \to \mathbb{R}^d$ over the L2-normalised output of a frozen backbone — linear with identity warm-start, or a 2-layer identity-residual MLP — trained to minimise L2-MSE between predicted and observed target latents on stable pairs. Across nine experiments we vary predictor class, optimiser, and backbone independently while holding every other knob constant.

This paper is a preliminary report. The single positive result we observe (frozen DermLIP at 0.944 ± 0.003 test AUROC across 5 seeds) *cannot be attributed to dermoscopy-domain transfer with the experiments reported here.* DermLIP's pretraining corpus, Derm1M (Yan et al., 2025), almost certainly contains HAM10000 raw images, and the BiomedCLIP partition we report rules out only the narrower "PMC-15M-style biomedical-figure pretraining (with PMC-15M's undocumented dermatology fraction) is sufficient" alternative; it does not preclude other medical pretraining corpora with substantial dermatology content. A natural follow-up experiment we sketch as EXP-009 (a self-pretrained DINOv2 on a non-HAM10000 dermoscopy corpus, §8) would aim to close the partition; we deliberately publish at this stage — before committing to that build — to gather community feedback on the methodology and the partition design.

Figure 1 previews the headline. Test AUROC on `strong_held_out_2` is below random for every frozen natural-image configuration we ran (four cells spanning two backbones, two predictor scaffolds, and two optimisers); barely lifts under a frozen general- medical backbone (BiomedCLIP, 0.329 ± 0.012); and reaches 0.944 ± 0.003 under a frozen dermoscopy-specific backbone (DermLIP), with the contamination caveat marked on the right.

Contributions.

- **Methodological.** A leakage-controlled HAM10000 longitudinal-proxy task with three disjoint synthetic nuisance families and a `train-on-two/evaluate-on-third` protocol, released as code, configurations, and reproduction launchers.
- **Empirical (negative).** Across four frozen-natural-image configurations (DINOv2 / CLIP $\times$ linear / MLP $\times$ SGD / Adam, with the four cells listed in Appendix A), test AUROC on the held-out third family inverts below random.
- **Empirical (positive but unpartitioned).** Frozen DermLIP reaches test AUROC = 0.944 ± 0.003 across 5 seeds, +0.363 above the strongest baseline. The win cannot yet be attributed to dermoscopy- domain transfer rather than HAM10000 image-level overlap; EXP-009 (§8) would be one way to close the partition.
- **Empirical (negative-as-partition).** Frozen BiomedCLIP — a biomedical-figure-pretrained backbone with no documented raw HAM10000/ISIC archive ingestion, and with an undocumented but plausibly small dermatology fraction — lifts only +0.04 over web CLIP, narrowing the open question to dermoscopy-rich pretraining.
- **Reproducibility.** Every primary-tier run is archived publicly at `abdelstark/derma-jepa-runs` with manifests, embeddings, metrics, model card, and logs. A single launcher

script reproduces each run end-to-end on a free A10G via Hugging Face Jobs.

- **A clearly scoped open question.** A candidate partition experiment (EXP-009) and the decision table that would attach to it are stated up front (§8) rather than treated as future work in passing.

2 Related work

JEPA-style and self-supervised image representations. The Joint-Embedding Predictive Architecture family (LeCun, 2022; Assran et al., 2023) predicts in representation space rather than pixel space, motivated by the conjecture that nuisance-invariant representations emerge from learning to predict the latents of related views. DINO (Caron et al., 2021) and DINOv2 (Oquab et al., 2024) train transformers via self-distillation to produce features that transfer across image distributions and downstream tasks; CLIP (Radford et al., 2021) and OpenCLIP (Cherti et al., 2023) build transferable representations via contrastive image–text alignment; MAE (He et al., 2022) reconstructs masked pixels and VICReg (Bardes et al., 2022) regularises variance, invariance, and covariance of the embedding. We read all of these as different routes toward representations that are approximately invariant to many photometric and framing nuisances; our probe asks whether that invariance generalises to nuisance families held out from the probe's training distribution. Our probe is narrow: the frozen-backbone, single-step, stable-pair subset of the JEPA family applied as an evaluation probe, not a pretraining objective.

Domain-specific medical foundation models. BiomedCLIP (Zhang et al., 2023) is a CLIP-style model trained on PMC-15M, ∼15 M biomedical image–text pairs from ∼4.4 M scientific articles. DermLIP (Yan et al., 2025) is a CLIP-style model trained on Derm1M, a ∼1 million dermatology image–text corpus that aggregates publicly available dermoscopy datasets and dermatology-website content. A broader dermatology foundation-model line includes MONET (Kim et al., 2024) (image–text alignment over dermatology textbook figures and PubMed) and PanDerm (Yan et al., 2024) (a multi-modal self-supervised model trained on ∼2M dermatology images); RETFound (Zhou et al., 2023) is a comparable medical-domain foundation-model precedent in retinal imaging, cited here for the domain-specific FM design pattern rather than as a methodological template for our probe. We use BiomedCLIP and DermLIP as third-party frozen weights, not as artefacts to evaluate in their own right; the paper makes no claim about either model beyond the test AUROC each produces under our specific probe.

Distribution-shift evaluation. WILDS (Koh et al., 2021) and the in-search-of-lost-DG framing (Gulrajani and Lopez-Paz, 2021) establish the modern protocol for distribution-shift evaluation. Our contribution is complementary: a deterministic, leakage- controlled, three-family-disjoint nuisance design that lets the experimenter precisely control the operation-level overlap between train and test nuisance. Where WILDS varies natural distributions (hospital, scanner, country), we vary synthetic nuisance families partitioned along the augmentation axis. The two designs answer different questions; the methodology proposed here is most useful when the natural axis of variation is hard to disentangle from the target signal, as in dermoscopic re-photography. ISIC challenge artefacts (Codella et al., 2019; Combalia et al., 2019) provide additional dermoscopy corpora that remain unprobed in this work and are candidates for the EXP-009 partition pretraining set (§8).

3 Method

3.1 Notation and probe

Let $\mathcal{X} \subset [0,1]^{3\times H\times W}$ denote RGB images at the backbone's native input resolution, and let $f : \mathcal{X} \to \mathbb{R}^d$ be a frozen vision encoder with output dimension d. Define the L2-normalised representation

$$z(x) \;=\; \frac{f(x)}{\|f(x)\|_2} \;\in\; \mathbb{S}^{d-1}. \tag{1}$$

A *stable* pair (x_c, x_t) shares a single underlying lesion image u subjected to two independent draws of a synthetic nuisance augmentation $\nu \sim \mathcal{N}_F$ from family F:

$$x_c = \nu_c(u), \qquad x_t = \nu_t(u), \qquad \nu_c, \nu_t \sim \mathcal{N}_F. \tag{2}$$

A *changing* pair pairs two distinct lesions u_1, u_2 matched on diagnosis $\mathrm{dx}(u_1) = \mathrm{dx}(u_2)$ and anatomical site $\mathrm{site}(u_1) = \mathrm{site}(u_2)$; each side receives an independent nuisance draw. The predictor $g_\theta : \mathbb{S}^{d-1} \to \mathbb{R}^d$ takes a unit-norm backbone representation as input and produces an unconstrained d-vector as output; predictor outputs are L2-normalised only at scoring time (eq. 5). It minimises the L2-MSE objective on stable training pairs:

$$\mathcal{L}(\theta) \;=\; \frac{1}{|\mathcal{D}^{\text{stable}}_{\text{train}}|} \sum_{(x_c,x_t)\in\mathcal{D}^{\text{stable}}_{\text{train}}} \|g_\theta(z(x_c)) - z(x_t)\|_2^2 \;+\; \mathcal{R}(\theta), \tag{3}$$

where $\mathcal{R}$ is a predictor-specific regulariser. For the linear predictor we use a soft `weight − I` penalty, $\mathcal{R}(\theta) = \lambda\,\|W - I\|_F^2$, which, in combination with the identity warm-start ($W \approx I$ at step zero, §3.2), holds the function near the identity at initialisation. For the 2-layer identity-residual MLP, the residual weight W_2 is zero-initialised so the predictor is identity at step zero by construction, and $\mathcal{R}$ is standard L^2 weight decay on the predictor weights (decoupled, in the Adam configuration); applying a $\|W_2 - I\|_F^2$ penalty here would contradict the zero-init identity warm-start. Per-variant values of λ are listed in Appendix B.

We do not L2-project predictions during training: expanding the loss, $\|g_\theta(z(x_c)) - z(x_t)\|_2^2 = \|g_\theta(z(x_c))\|_2^2 - 2\,\langle g_\theta(z(x_c)), z(x_t)\rangle + 1$, so the loss simultaneously drives the prediction direction toward $z(x_t)$ and the prediction norm toward 1. Empirically the prediction-norm mean is ≥ 0.97 in every reported run (Appendix E). The $\|W - I\|_F^2$ penalty is soft, not constraining: at equilibrium $\|W - I\|_F$ is set by the balance between the data gradient and λ, not pinned at zero.

At evaluation time, the score for a pair is the cosine *distance* between the L2-normalised prediction and the L2-normalised target:

$$\tilde{g}_\theta(z) \;:=\; \frac{g_\theta(z)}{\|g_\theta(z)\|_2}, \tag{4}$$

$$s(x_c, x_t) \;=\; 1 \;-\; \langle \tilde{g}_\theta(z(x_c)), z(x_t)\rangle. \tag{5}$$

We adopt $y = 1$ for changing and $y = 0$ for stable, and use s directly (no sign flip) as the score input to AUROC, so that a correctly oriented predictor produces $s(x_c, x_t) < s(x'_c, x'_t)$ whenever (x_c, x_t) is stable and (x'_c, x'_t) is changing, and AUROC > 0.5. Under this fixed convention, AUROC < 0.5 is a structurally inverted predictor, not a sign-convention free parameter: in deployment the predictor cannot flip its output without an oracle that already knows which class is which, so the directional AUROC is the relevant quantity.

3.2 Predictor variants

Linear with identity warm-start. $g_\theta(z) = Wz + b$ with W initialised to $I + \epsilon$, $\epsilon_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.005^2)$, and $b = 0$. Under SGD with the soft `weight − I` penalty above, the predictor at step zero is identity-equivalent.

2-layer identity-residual MLP. $g_\theta(z) = z + W_2\,\mathrm{ReLU}(W_1 z + b_1) + b_2$, with W_2 zero-initialised so the residual term vanishes at step zero and $g_\theta =$ identity. Hidden dimension 512.

3.3 Three nuisance families

The augmentation suite specifies three deterministic synthetic nuisance families, summarised in Table 1. No operation–parameter-band combination is shared between any two families; JPEG compression is the one operation that recurs across all three, with disjoint quality bands (45–70 for `strong`, 20–40 for `strong_held_out`, 80–95 for `strong_held_out_2`) chosen so the resulting compression artefact is perceptually distinct per family. The exact operation parameters are listed in Appendix C.

Family	Operation set
`strong`	brightness × contrast × saturation, rotation ±15°, scale 0.82–1.00, translate ±5%, horizontal flip, Gaussian blur, Gaussian noise, JPEG quality 45–70
`strong_held_out`	hue shift ±12° (HSV rotation), posterise 4–6 bits, unsharp-mask sharpen, motion blur (linear kernel 5–15 px), random rectangular erasing 6–18% area, JPEG quality 20–40
`strong_held_out_2`	gamma 0.6–1.6, colour temperature 0.85–1.15, radial vignette 25–55% falloff, salt-and-pepper noise 0.5–2.5%, JPEG quality 80–95

Table 1: Three deterministic nuisance families, partitioned along the operation axis. Train stable pairs draw from {`strong`, `strong_held_out`}; val and test stable pairs use `strong_held_out_2`.

3.4 Splits and pair construction

We use lesion-ID-aware splits at fractions 0.70/0.15/0.15, deterministic from a fixed seed, with group counts 5,229/1,120/1,121 across HAM10000's 10,015 images and 7,470 unique lesions. Each split contains 1,000 stable plus 1,000 changing pairs (total 6,000 pairs per run). The strict same-diagnosis-site changing-pair policy requires that a context lesion's changing partner be a distinct lesion with identical diagnosis and identical anatomical site. Train stable pairs rotate deterministically between `strong` and `strong_held_out` by pair index; val and test stable pairs use `strong_held_out_2` exclusively.

3.5 Evaluation protocol

We report the area under the receiver-operating-characteristic curve (AUROC) as the primary metric, with 1,000-sample bootstrap 95% confidence intervals computed by per-pair resampling. The area under the precision–recall curve (AUPRC), the equal-error-rate (EER) threshold, and the false-positive rate (FPR) at fixed true-positive rate (TPR) = 0.8 accompany every run, as do three representation- health checks: prediction-norm mean and minimum, dimension-variance mean and minimum, and a binary `collapsed` flag triggered when either falls below 10^{-8}. None of the nine reported runs is flagged as collapsed. Section 4 fixes the backbones, training hyperparameters, and compute envelope used to instantiate this protocol.

4 Experimental setup

Backbones. Five frozen vision backbones cover the architectural and pretraining-data axes the paper varies (Table 2).

Backbone	Arch.	d	Pretraining	Hugging Face Hub path
DINOv2 ViT-S/14	ViT-S/14	384	LVD-142M (web)	`facebook/dinov2-small`
DINOv2 ViT-B/14	ViT-B/14	768	LVD-142M (web)	`facebook/dinov2-base`
OpenAI CLIP ViT-B/16	ViT-B/16	512	WIT (web)	`openai/clip-vit-base-patch16`
DermLIP ViT-B/16	ViT-B/16	512	Derm1M (derm.)	`redlessone/DermLIP_ViT-B-16`
BiomedCLIP ViT-B/16	ViT-B/16	512	PMC-15M (med.)	`microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224`

Table 2: Frozen vision backbones evaluated. Citations: DINOv2 (Oquab et al., 2024), OpenAI CLIP (Radford et al., 2021), DermLIP (Yan et al., 2025), BiomedCLIP (Zhang et al., 2023). Hugging Face (HF) Hub paths resolve at `https://huggingface.co/`⟨path⟩.

Training. 200 epochs at batch size 128 on 1,000 training pairs. The linear predictor uses SGD at LR 3×10^{-2} with weight decay 10^{-3} applied to `(weight − I)`. The MLP predictor uses Adam at LR 10^{-3} with decoupled weight decay 10^{-4}. Identity warm-start in both cases.

Compute. Each primary-tier run executes on a single NVIDIA A10G 24 GB via Hugging Face Jobs in approximately 95–100 minutes wall-clock end-to-end (manifest build, embedding export, baseline evaluation, predictor training, and run-archive upload). Hosted dependency versions are pinned in `scripts/hf_jobs_constraints.txt`.

Code, configurations, and run archive. Source code is at `https://github.com/AbdelStark/derma-jepa`. Per-experiment configurations and launchers live under `configs/data/` and `scripts/`, one pair per experiment. The run archive containing every primary-tier run's manifests, embeddings, metrics, model card, and logs is at `abdelstark/derma-jepa-runs`.

5 Experimental sequence

We report the nine experiments as a falsification ladder, each isolating one axis the previous experiment left ambiguous. Subsection headings give the run identifier; the corresponding self-contained report under `docs/experiments/EXP-NNN-...md` carries the full breakdown. Numbers in this section trace to those reports' §4.1.

EXP-001 — sanity. A trivial proxy with mild nuisance produces ceiling AUROC ≈ 1.000 across every dense baseline (pixel L2, SSIM, DINOv2 cosine). The proxy needs hardening.

EXP-002 — hardened proxy, matched eval. With `strong` nuisance applied to both training and evaluation pairs, a linear JEPA-style predictor over frozen DINOv2 ViT-B/14 reaches test AUROC = 0.920, +0.269 above the strongest baseline (DINOv2 ViT-S/14 cosine = 0.652).

EXP-003 — one-family-held-out evaluation. Training stable pairs use `strong`, evaluation pairs use `strong_held_out`. Test AUROC falls to 0.680, −0.281 below the strongest baseline (SSIM distance = 0.961). The matched-eval delta does not transfer to a single disjoint nuisance family.

EXP-004 — mixed-family training, third-family evaluation. Training stable pairs rotate between `strong` and `strong_held_out`; evaluation uses the third disjoint family `strong_held_out_2`. Test AUROC inverts to 0.249, −0.331 below the strongest baseline (pixel L2 = 0.580). The proxy is now decisively harder than EXP-001 and the predictor's extrapolation to the unseen family produces a systematically inverted ranking, not noise.

EXP-005, EXP-006a — predictor-class ablation. Replacing the linear predictor with a 2-layer identity-residual MLP under SGD (EXP-005) underfits training (train AUROC = 0.572, test AUROC = 0.270). Switching to Adam at LR 10^{-3} (EXP-006a) lets the MLP fit (train AUROC = 0.893) but produces test AUROC = 0.248, within a single bootstrap CI of EXP-004 linear's 0.249. The scaffold-capacity hypothesis is not supported by these data.

EXP-006b — backbone ablation, natural-image. Replacing DINOv2 ViT-B/14 with OpenAI CLIP ViT-B/16, holding everything else identical, produces test AUROC = 0.286. The inversion is not DINOv2-specific.

EXP-007 — pretraining-domain ablation. Replacing OpenAI CLIP with DermLIP — the same ViT-B/16 architecture CLIP-trained on Derm1M dermatology image–text pairs instead of OpenAI WIT web image–text pairs — produces test AUROC = 0.945 on the original seed (5-seed mean 0.944 ± 0.003, see seed-sweep paragraph below). The train→test drop collapses from −0.70 (EXP-006b) to −0.05. *This is the only above-random result on `strong_held_out_2` across the nine runs.* We introduce the contamination caveat here and pursue it in EXP-008.

EXP-008 — pretraining-domain partition. Replacing DermLIP with BiomedCLIP — ViT-B/16 CLIP-trained on PMC-15M biomedical figure–caption pairs, with no documented raw HAM10000/ISIC archive ingestion — produces test AUROC = 0.325 (single seed). The web → general-medical step adds +0.04 AUROC; the general-medical → dermoscopy step adds +0.62. EXP-008 rules out "PMC-15M-style biomedical-figure pretraining at the dermatology fraction PMC-15M happens to contain" as sufficient (PMC-15M's per-source breakdown is not published, so we do not generalise beyond the PMC-15M-shaped point) but does not separate dermoscopy-domain transfer from HAM10000 image-level overlap.

Seed sweep on EXP-007 and EXP-008. Five seeds per configuration. DermLIP test AUROC = 0.944 ± 0.003 (range 0.939–0.947, 95% CI[mean] $[0.941, 0.946]$); BiomedCLIP test AUROC = 0.329 ± 0.012 (range 0.312–0.344, 95% CI[mean] $[0.318, 0.339]$). Both headlines are seed-stable; the partition gradient survives seed variance.

6 Results

Table 3 consolidates the nine-run sequence under the EXP-004 proxy. Figure 1 visualises the pretraining-data gradient. Per-run plots — pair-score histograms, training-loss trajectories, and representation-health values — live in the run archive at abdelstark/derma-jepa-runs under each run's `<run_id>/` folder; the appendix reproduces the most relevant cross-run summary tables.

Table 4 isolates the predictor × optimiser axis under frozen DINOv2 ViT-B/14: across linear and MLP predictors, under SGD and Adam, with train AUROC ranging from 0.572 to 0.900, test AUROC on `strong_held_out_2` stays in $[0.248, 0.270]$ (a ± 0.011 half-range around the midpoint 0.259). Scaffold capacity is not the bottleneck.

Table 5 reports the seed sweep on the two contamination-relevant configurations. Both headline numbers are seed-stable; the partition gradient is robust to seed variance.

Figure 2 plots train versus test AUROC for each primary-tier run. The DINOv2 and CLIP runs cluster in the top-right / bottom-right region (high train, low test) — a sharp generalisation gap. EXP-005 sits low on both axes (the under-fit MLP that barely departed from identity). The DermLIP point is the only one on the diagonal; BiomedCLIP fits training but does not transfer.

Figure 3 shows the test-split pair-score distributions for the three CLIP-architecture backbones. The predictor's score is the cosine distance between its predicted target and the observed

Run	Backbone	Predictor	Opt.	Train AUROC	Test AUROC	Δ
EXP-001	DINOv2 ViT-B/14	linear	SGD	1.000	1.000	0.000
EXP-002	DINOv2 ViT-B/14	linear	SGD	0.953	0.920	+0.269
EXP-003	DINOv2 ViT-B/14	linear	SGD	0.953	0.680	−0.281
EXP-004	DINOv2 ViT-B/14	linear	SGD	0.900	0.249	−0.331
EXP-005	DINOv2 ViT-B/14	MLP	SGD	0.572	0.270	−0.310
EXP-006a	DINOv2 ViT-B/14	MLP	Adam	0.893	0.248	−0.332
EXP-006b	OpenAI CLIP B/16	linear	SGD	0.986	0.286	−0.294
EXP-008	BiomedCLIP B/16	linear	SGD	$0.912^{\dagger}$	$\mathbf{0.329 \pm 0.012}^{\ddagger}$	−0.252
EXP-007	DermLIP B/16*	linear	SGD	1.000	$\mathbf{0.944 \pm 0.003}^{\ddagger}$	**+0.363**

Table 3: Cross-run results on the EXP-004 proxy (HAM10000, leakage-controlled splits, 1,000 stable + 1,000 changing pairs per split). Test column is AUROC on `strong_held_out_2`; Δ is versus the strongest cheap baseline on each split (pixel L2 = 0.580 for EXP-004 onward; SSIM or DINOv2-S cosine in earlier runs). † Seed-mean of the BiomedCLIP train AUROC across five seeds. ‡ Mean ± sd across five seeds. * Derm1M almost certainly contains HAM10000 raw images; the resulting positive cannot be attributed to dermoscopy-domain transfer with the experiments reported here (§8). AUROC is reported under the directional convention of §3.5 ($y = 1$ for changing, no sign flip); the sign-symmetric equivalent for raw embedding cosine baselines (whose discriminative information survives a sign flip but is not usable in deployment without an oracle that already labels the pair) is $1 - \text{AUROC}$, e.g. symmetric raw DermLIP cosine = 0.891, raw OpenAI CLIP cosine = 0.964, raw BiomedCLIP cosine = 0.953, raw DINOv2 ViT-B/14 cosine = 0.726. The directional comparison is the operative one because the predictor must learn the correct orientation under the fixed train/test convention.

Run	Predictor	Opt.	Train AUROC	Test AUROC	95% CI
EXP-004	linear (identity warm-start)	SGD	0.900	0.249	$[0.230, 0.270]$
EXP-005	ID-residual MLP (under-fit, W_2 tiny)	SGD	0.572	0.270	$[0.249, 0.293]$
EXP-006a	ID-residual MLP (fit)	Adam	0.893	0.248	$[0.228, 0.269]$

Table 4: Predictor × optimiser ablation under frozen DINOv2 ViT-B/14 on the EXP-004 proxy. Across two predictor classes and two optimisers, train AUROC ranges from 0.572 (under-fit MLP) to 0.900 (linear) but test AUROC on `strong_held_out_2` stays in a interval $[0.248, 0.270]$ (half-range 0.011 around the midpoint 0.259). Scaffold capacity does not move the held-out-family floor.

target, so a correctly oriented predictor places *stable*-pair scores below *changing*-pair scores. CLIP and BiomedCLIP exhibit the inverted ordering — stable mean above changing mean by 0.049 and 0.034 respectively — while DermLIP separates them by +0.111 in the correct direction. The structural inversion visible here is the mechanism that holds test AUROC to 0.25–0.33 across natural-image and general-medical pretraining rather than at chance.

7 Analysis and discussion

7.1 Why the inversion is below random rather than at chance

The structured-inversion finding from §6 is not noise: the linear predictor learns a direction $w \in \mathbb{R}^d$ that encodes "stable under $\{\texttt{strong}, \texttt{strong\_held\_out}\}$" as small $\langle w, z_t - z_c \rangle$, and this direction extrapolates with the *opposite* sign when the unseen family `strong_held_out_2` rotates the latent geometry the other way. Because the third family's nuisance is operation-disjoint from the training families, the rotation does not partially align — it anti-aligns. Figure 3 shows this structure directly: under frozen OpenAI CLIP, the test-split stable-pair mean cosine distance

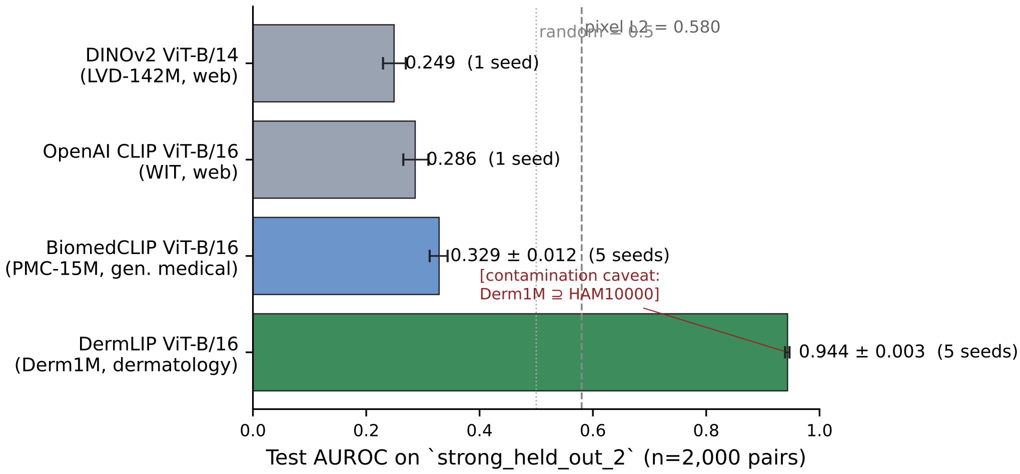

Figure 1: Test AUROC on `strong_held_out_2` (the unseen third nuisance family) under the EXP-004 recipe, across four backbones. Error bars are bootstrap 95% CIs (single seed) or seed-mean ± standard deviation across five seeds where available. Pixel L2 (the strongest cheap baseline at 0.580) is shown as a dashed reference line. The DermLIP bar carries the contamination caveat: Derm1M almost certainly contains HAM10000, and the experiments reported here cannot separate dermoscopy-domain transfer from HAM10000 image-level overlap (§8).

Run	Backbone	n seeds	Test AUROC (mean ± sd)	Range	95% CI[mean]
EXP-007	DermLIP B/16	5	0.9435 ± 0.0029	$[0.939, 0.947]$	$[0.941, 0.946]$
EXP-008	BiomedCLIP B/16	5	0.3286 ± 0.0120	$[0.312, 0.344]$	$[0.318, 0.339]$

Table 5: Seed sweep on EXP-007 and EXP-008. Five seeds per configuration. Both headlines are seed-stable; the seed-mean pretraining-data gradient (web 0.286 < general-medical 0.329 < dermoscopy 0.944) survives. The $\sim 4\times$ ratio between the two seed standard deviations is consistent with the high-AUROC DermLIP regime saturating predictor parameters more reproducibly than the partial-fit BiomedCLIP regime.

(0.281) sits *above* the changing-pair mean (0.233), a -0.049 inversion gap. Under BiomedCLIP, the gap narrows to -0.034. Under DermLIP, the ordering flips to the correct direction with a $+0.111$ gap (stable 0.236, changing 0.346). The natural-image inversion is large enough relative to the within-class spread that AUROC sits in the 0.25–0.29 band rather than at chance.

7.2 Why DermLIP transfers under this scaffold and BiomedCLIP doesn't

The +0.66 AUROC swing from EXP-006b (OpenAI CLIP) to EXP-007 (DermLIP) cannot be explained by architecture, scaffold, or optimiser: all three are held byte-identical. The plausible mechanistic accounts are

(i) DermLIP's CLIP loss over $\sim$1 M dermatology image–text pairs has organised the embedding space such that a single linear direction captures stable-pair similarity across all three nuisance families uniformly, including `strong_held_out_2`; or

(ii) DermLIP saw the HAM10000 raw images during pretraining and its embedding space is well-fit to HAM10000 lesion identity in a way that survives the synthetic post-hoc nuisance.

EXP-008 rules out a third candidate — "PMC-15M-style biomedical- figure pretraining is sufficient" — by showing that BiomedCLIP, trained on $\sim$15 M PMC-15M biomedical image–text pairs with no documented raw HAM10000/ISIC archive ingestion, lifts test AUROC by only +0.04 over web CLIP. PMC-15M's dermatology fraction is undocumented, so the negative does

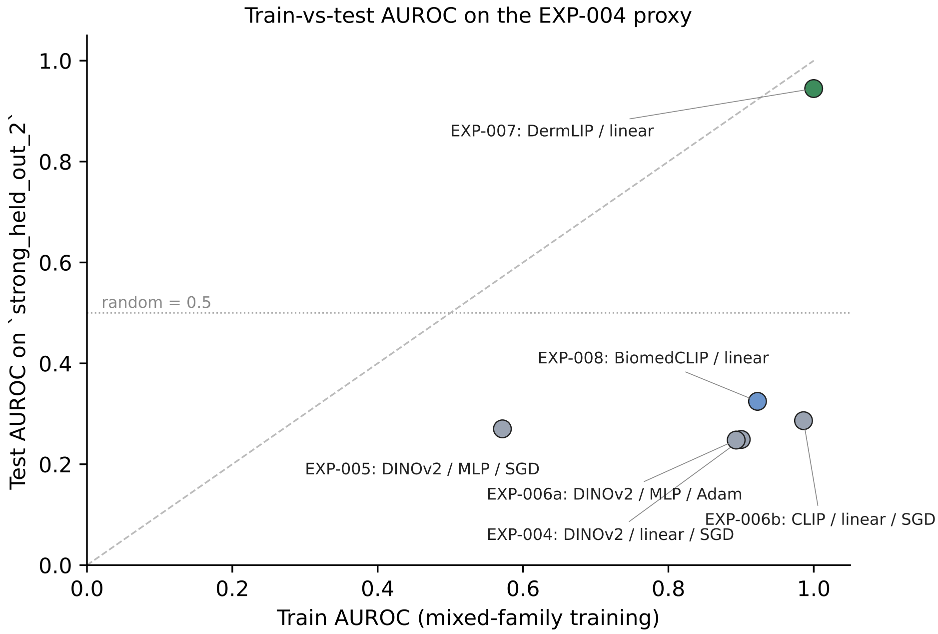

Figure 2: Train versus test AUROC for each primary-tier run on the EXP-004 proxy. The dashed diagonal is train = test; the dotted horizontal is the random baseline at 0.5. DermLIP is the only configuration whose test AUROC tracks training; the four natural-image configurations sit in the inverted band below the random line.

not generalise to all "general medical" pretraining; a corpus with substantial dermoscopy content but no HAM10000 (which EXP-009 below would assemble) remains the open partition. EXP-008 cannot, however, choose between (i) and (ii). EXP-009 (a non-HAM10000 dermoscopy SSL pretrain, §8) is the experiment that would.

7.3 Loss versus AUROC

MSE loss delta does not predict train AUROC across optimisers. From the per-run training-dynamics tables in the experiment reports (`docs/experiments/EXP-{004,006a}-...md` §4.3), EXP-006a (Adam, MLP) reduces train MSE by 3.3% over 200 epochs (epoch-1 loss 6.94×10^{-4} $\rightarrow$ epoch-200 loss 6.71×10^{-4}) and reaches train AUROC = 0.893. EXP-004 (SGD, linear) reduces train MSE by 12.6% ($7.87\times10^{-4} \rightarrow 6.88\times10^{-4}$) and reaches train AUROC = 0.900. The optimisers move the predictor's function very differently per unit of L2-loss reduction in 768-d normalised space. We therefore report both metrics; headline-AUROC comparisons should not be replaced by loss-only comparisons.

7.4 What is not yet claimed

We do not claim that frozen DermLIP "solves" the proxy in a transfer sense. We claim that frozen DermLIP plus a linear scaffold reaches AUROC = 0.944 ± 0.003 on `strong_held_out_2` under the specific evaluation protocol of §3, with HAM10000 contamination unpartitioned. The result is consistent with two hypotheses — dermoscopy-domain transfer and HAM10000 image-level overlap during DermLIP pretraining — and EXP-009 is required to choose between them. The BiomedCLIP partition rules out only the broader "any medical-image pretraining is sufficient" alternative.

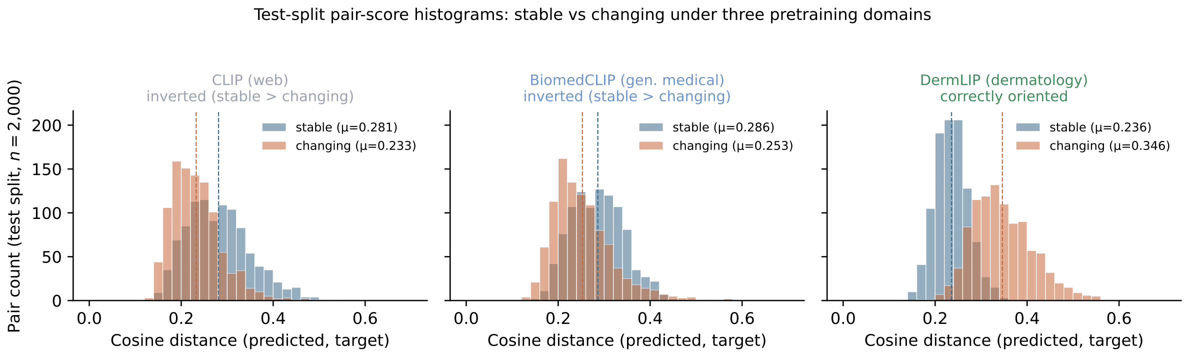

Figure 3: Test-split pair-score histograms (cosine distance between the predictor's predicted target and the observed target) under three pretraining-data domains, holding architecture (ViT-B/16) and predictor (linear) constant. Vertical dashed lines mark per-class means. Stable pairs should lie below changing pairs; the CLIP and BiomedCLIP panels show the inversion that pulls test AUROC below random, while the DermLIP panel separates the two classes by +0.111 in the correct direction.

8 Candidate follow-up: EXP-009

We sketch one natural candidate experiment that, if eventually run, would partition the two hypotheses behind the DermLIP positive. We present the design now to invite feedback on whether this is the right follow-up before we (or anyone else) invests the engineering effort required to run it.

8.1 The partition question

The DermLIP positive in EXP-007 (AUROC $= 0.944 \pm 0.003$) is consistent with two distinct hypotheses. Either DermLIP's dermoscopy-domain pretraining on Derm1M's non-HAM10000 fraction has aligned nuisance directions across the three families and that alignment generalises to `strong_held_out_2`, or DermLIP saw HAM10000 raw images during pretraining and the embedding space is well-fit to HAM10000 lesion identity in a way that survives our synthetic nuisance. EXP-008's BiomedCLIP partition rules out only the broader "any medical-image pretraining" alternative. The dermoscopy-vs-HAM10000 distinction remains open.

8.2 EXP-009 design

Self-pretrain a DINOv2 ViT-B/14 encoder on a non-HAM10000 dermoscopy corpus assembled from ISIC archives minus the HAM10000 split, DermNet, DermQuest, BCN20000 (Combalia et al., 2019) non-HAM10000 components, and similar publicly available dermoscopy and clinical-skin sources. Run a short JEPA-style or masked-image-modelling objective (target: ∼20 epochs, ∼200 k steps), then re-run the EXP-004 recipe on top with the linear predictor.

8.3 Decision table

Table 6 maps the EXP-009 test-AUROC band to a revision of the paper's claim.

8.4 Why we share the design before running it

Corpus assembly is multi-week: provenance audit on every ISIC component to confirm HAM10000 exclusion, SSL pretrain compute on a non-trivial GPU budget, and a pretrain-quality check on a held-out HAM10000 split. Releasing the methodology and the nine-run arc at this stage is intentional — the paper seeks community feedback on the proxy-task design

EXP-009 test AUROC	Interpretation	Effect on the paper claim
≥ 0.85	Dermoscopy-domain transfer is sufficient.	Headline survives; caveat downgrades to "verified."
0.50–0.80	Partial transfer; mixed contribution.	Claim becomes "domain helps, dataset overlap helps more."
0.30–0.50	HAM10000 overlap drove EXP-007.	Claim narrows to "frozen-backbone JEPA unlocks when the backbone has seen the eval dataset."

Table 6: Three EXP-009 outcome bands on `strong_held_out_2` and the corresponding revision of the paper's claim. Committing this mapping up front would prevent post-hoc re-interpretation of the experiment.

and on whether this partition is the right follow-up before anyone (including us) commits to the build. Whether EXP-009, a different partition, or no follow-up at all is the right next step depends on that feedback.

8.5 Feedback we are seeking

1. Is the proxy-task design (three deterministic nuisance families with operation-level disjointness, train-on-two / evaluate-on-third) a reasonable test of frozen-backbone generalisation, or is a different construction more informative?
2. Is the proposed partition (a non-HAM10000 dermoscopy SSL pretrain) the right one, or would a different partition (e.g. a dermoscopy-text-only retrieval probe over DermLIP, or a fine-tuning-on-HAM10000 ablation of a known-non-contaminated backbone) be more informative?
3. What other follow-ups would matter for the next stage — different architecture classes, real same-lesion-over-time data, or a different evaluation metric?
4. Are there pre-trained backbones in the public literature that would be cleaner partition points than the ones we have used?

9 Limitations

1. **Pretraining contamination is unpartitioned.** The DermLIP positive cannot be attributed to dermoscopy-domain transfer with the experiments reported here. EXP-009 (§8) is one candidate partition experiment we have sketched; until such a partition is run, the headline should be read as "frozen DermLIP plus a linear scaffold reaches AUROC = 0.944 under the conditions of §3, with HAM10000 contamination unpartitioned" rather than as a transfer claim.
2. **Synthetic nuisance.** The three families are deterministic operations on PIL images. Real-world dermoscopic acquisition variability is more complex; the proxy is methodological and is identified as such.
3. **Single-architecture-class comparison.** All four backbones in the cross-backbone gradient are ViT-B-class (CLIP architecture at patch size 16 for OpenAI CLIP, DermLIP, and BiomedCLIP; DINOv2 architecture at patch size 14 for DINOv2). Other architecture classes — different capacity tiers, SigLIP, hierarchical or convolutional backbones — are unprobed. Future work should vary architecture class as a separate axis.
4. **Single dataset.** HAM10000 is the only evaluation corpus. ISIC archive components and PAD-UFES-20 are unprobed in this work and are candidates for v2 experiments and for the EXP-009 corpus.
5. **HAM10000 is cross-sectional.** The dataset has no real same-lesion-over-time pairs; the

longitudinal proxy is synthetic throughout. Validating the methodology on a real longitudinal dataset (e.g. ISIC 2020's serial-imaging metadata) is an explicit next-stage follow-up.

6. **Single scaffold class on the dermoscopy backbone.** The MLP ablation (EXP-005, EXP-006a) was run only on DINOv2; an MLP-on-DermLIP run was not performed. Predicted to land at or above EXP-007's 0.944 since the linear scaffold already saturates training, but unverified.
7. **Unstratified bootstrap.** The pair-level bootstrap (Appendix D) is not stratified by label, diagnosis, or anatomical site, so the reported 95% CIs are marginal over the empirical class balance. Stratified resampling would tighten CIs slightly but is not expected to move headline numbers.

10 Conclusion

We have characterised when frozen-backbone JEPA-style probes fail and when they succeed under a leakage-controlled, three-family-disjoint nuisance design on HAM10000. Across four frozen-natural-image configurations spanning two backbones, two predictor scaffolds, and two optimisers (the four cells are listed in Appendix A; we did not run the full $2 \times 2 \times 2$ factorial), test AUROC on the held-out third nuisance family inverts below random. Frozen general- medical pretraining (BiomedCLIP, PMC-15M scientific-article figures, with no documented raw HAM10000/ISIC archive ingestion) lifts test AUROC by only +0.04. Frozen dermoscopy-pretrained DermLIP reaches test AUROC $= 0.944 \pm 0.003$ across 5 seeds — the only above-random result in nine runs — but under a pretraining corpus that almost certainly contains HAM10000. The methodology, the configurations, and the run archive are released. We do not claim to have validated or invalidated the underlying thesis at this stage; we have characterised a sharp failure mode of the natural-image and general-medical configurations, observed one positive configuration whose attribution is unresolved, and sketched one candidate partition experiment (EXP-009) that, if eventually run, could resolve it. Feedback on the proxy-task design and on whether this partition is the right next step is the central purpose of this preliminary release.

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A Per-run details

This appendix re-states the nine-run sequence and the ten-run seed sweep one entry at a time, anchored on the run identifier so that readers can navigate from a quoted number to the run-archive folder and back. Headline numbers therefore overlap with §5; the unique content here is the run-ID, the train→test gap, and any run-specific anomaly. Each run has a self-contained report at `docs/experiments/EXP-NNN-...md` in the source repository, and each run's `manifest.json`, `embeddings/`, `metrics.json`, `baseline_metrics.json`, `config.yaml`, `model_card.md`, and `logs/` subtree are mirrored under `abdelstark/derma-jepa-runs` at <run_id>/.

A.1 EXP-001 — ceiling sanity

`ham10000-hf-dinov2-primary-v1`. DINOv2 ViT-B/14, linear, SGD; mild matched-family nuisance. Train/val/test AUROC $\approx$ 1.000 across every dense baseline and the predictor. Diagnoses the proxy as too easy and motivates the harder constructions in EXP-002 onward.

A.2 EXP-002 — hardened proxy, matched evaluation

`ham10000-hf-dinov2-exp002-v1`. `strong` on both train and test splits. Linear-on-DINOv2 reaches test AUROC = 0.920, +0.269 above DINOv2 ViT-S/14 raw cosine (0.652) with non-overlapping bootstrap CIs. The "predict-in-latent-space beats cheap baselines" claim holds when the test nuisance distribution matches training.

A.3 EXP-003 — one-family-held-out evaluation

`ham10000-hf-dinov2-exp003-v1`. Train on `strong`, evaluate on `strong_held_out`. Test AUROC falls to 0.680; SSIM distance becomes the strongest baseline at 0.961. The matched-eval delta does not transfer to a single disjoint nuisance family.

A.4 EXP-004 — mixed-family training, third-family evaluation

`ham10000-hf-dinov2-exp004-v1`. Train stable pairs rotate between `strong` and `strong_held_out`; evaluation on `strong_held_out_2`. Test AUROC = 0.249 $[0.230, 0.270]$; below-random inversion. Pixel L2 (0.580) is the strongest baseline. This run defines the canonical "EXP-004 proxy" used by every subsequent backbone-ablation run.

A.5 EXP-005 — 2-layer identity-residual MLP under SGD

`ham10000-hf-dinov2-exp005-v1`. Same proxy as EXP-004; replace the linear predictor with the MLP variant under EXP-002's SGD recipe (LR 3×10^{-2}, weight decay 10^{-3}). Train AUROC = 0.572 (under-fit; the MLP departs only mildly from identity), test AUROC = 0.270. The scaffold-capacity hypothesis is partially tested but the MLP has barely moved off the identity init.

A.6 EXP-006a — MLP under Adam

`ham10000-hf-dinov2-exp006a-v1`. Same MLP as EXP-005 but Adam at LR 10^{-3}, decoupled weight decay 10^{-4}. The optimiser change lets the MLP fit: train AUROC = 0.893, close to EXP-004 linear's 0.900. Test AUROC = 0.248 $[0.228, 0.269]$ — within a single bootstrap CI of EXP-004 linear's 0.249 and below the raw DINOv2 ViT-B/14 cosine baseline of 0.274. The properly-fit MLP does not move the held-out-family test AUROC off the linear-predictor value.

A.7 EXP-006b — OpenAI CLIP ViT-B/16

`ham10000-hf-clip-exp006b-v1`. EXP-004 recipe with the backbone swapped from DINOv2 ViT-B/14 to OpenAI CLIP ViT-B/16. Test AUROC = 0.286 $[0.265, 0.310]$, overlapping bootstrap CIs with EXP-004's 0.249. The below-random inversion is not DINOv2-specific. Train AUROC = 0.986; the train→test drop is −0.70.

A.8 EXP-007 — DermLIP ViT-B/16

`ham10000-hf-dermlip-exp007-v1`. EXP-004 recipe with the backbone swapped from OpenAI CLIP to DermLIP. Test AUROC = 0.945 $[0.935, 0.954]$ on the original seed; first above-random result on `strong_held_out_2` since EXP-002 (and the only one across nine primary-tier runs). The contamination caveat is introduced here: Derm1M's data card does not publish a per-source breakdown, but HAM10000 raw images are almost certainly included (Appendix H).

A.9 EXP-008 — BiomedCLIP ViT-B/16

`ham10000-hf-biomedclip-exp008-v1`. EXP-004 recipe with the backbone swapped to BiomedCLIP (PMC-15M scientific-article figures, with no documented raw HAM10000/ISIC archive ingestion). Test AUROC = 0.325 $[0.303, 0.349]$. The web → general-medical step contributes +0.04; the general-medical → dermoscopy step contributes +0.62. EXP-008 rules out "any medical-image pretraining" as sufficient but does not partition dermoscopy-domain transfer from HAM10000 image-level overlap.

A.10 Seed sweep on EXP-007 and EXP-008

Five seeds per configuration, ten runs total. `...-dermlip-exp007-seed-{1,2,3,4}-v1` and `...-biomedclip-exp008-seed-{1,2,3,4}-v1` alongside the canonical seed-20260422 runs. DermLIP test AUROC = 0.9435 ± 0.0029 (range 0.9392–0.9470, 95% CI[mean] $[0.9409, 0.9461]$); BiomedCLIP test AUROC = 0.3286 ± 0.0120 (range 0.3119–0.3436, 95% CI[mean] $[0.3181, 0.3391]$). The seed-to-seed sd ratio ($0.012/0.003 \approx 4\times$) is consistent with the high-AUROC DermLIP regime saturating predictor parameters more reproducibly than the partial-fit BiomedCLIP regime. The pretraining-data gradient (web 0.286 < general-medical 0.329 < dermoscopy 0.944) survives seed variance.

B Hyperparameters

Table 7 lists the predictor hyperparameters used in every primary-tier run. Values are stable across all nine runs (and the ten seed-sweep runs); only the predictor type, the optimiser, and the backbone are varied between runs. The full per-run configurations are committed under `configs/data/` and mirrored alongside every run as `<run_id>/config.yaml`.

Embedding precondition. Backbone outputs are L2-normalised before entering the predictor (eq. 3). Predictor outputs are L2-normalised before the cosine-distance score (eq. 5). No batch normalisation, dropout, or stochastic depth is used inside the predictor.

Setting	Linear predictor	2-layer identity-residual MLP
Architecture	$g_\theta(z) = Wz + b$	$g_\theta(z) = z + W_2\,\mathrm{ReLU}(W_1 z + b_1) + b_2$
Hidden dim	—	512
Init W / W_1	$I + \epsilon$, $\epsilon_{ij} \sim \mathcal{N}(0, 0.005^2)$	Kaiming-uniform
Init W_2	—	zero ($g_\theta = \mathrm{id}$ at step 0)
Init biases	zero	zero
Optimiser	SGD	SGD (EXP-005) / Adam (EXP-006a)
Learning rate	3×10^{-2}	3×10^{-2} / 10^{-3}
Weight decay	10^{-3} on $(W - I)$	10^{-3} on W_1, W_2 / 10^{-4} decoupled
Batch size	128	128
Epochs	200	200
Grad. clipping	none	none
LR schedule	constant	constant
Loss	L2-MSE on stable pairs (eq. 3)	L2-MSE on stable pairs
Bootstrap	1,000 resamples, per-pair	1,000 resamples, per-pair
Seed (default)	20260422	20260422

Table 7: Predictor hyperparameters. Every other knob (split fractions, seed, pair counts, augmentation parameters) is held constant across all nine primary-tier runs and across the ten seed-sweep runs.

Backbone configurations. All backbones run in inference mode (`eval()`, no gradient). Embedding dimensions: DINOv2 ViT-B/14 $d = 768$; OpenAI CLIP ViT-B/16 $d = 512$; DermLIP ViT-B/16 $d = 512$; BiomedCLIP ViT-B/16 $d = 512$. The predictor input/output dimensionality matches the backbone output and is set automatically from each backbone's configured embedding size.

C Nuisance-family operation specifications

The three nuisance families in §3.3 are implemented in `src/derma_jepa/public_data.py`: `strong` in `_apply_strong_nuisance`, `strong_held_out` in `_apply_strong_held_out_nuisance`, and `strong_held_out_2` in `_apply_strong_held_out_2_nuisance`. Each function takes a PIL image, a per-pair index, and a NumPy `Generator`, applies the operations below in the listed order, and returns the nuisance-perturbed image alongside a recipe dictionary recording the sampled parameter values.

`strong`

Photometric, geometric, and sensor-noise perturbations that model ordinary acquisition variability.

- Brightness, contrast, and saturation jitter applied independently: each scaled by $u \sim \mathrm{Uniform}(0.6, 1.4)$ via PIL `ImageEnhance`.
- Affine transform: rotation $\theta \sim \mathrm{Uniform}(-15°, 15°)$, isotropic scale $s \sim \mathrm{Uniform}(0.82, 1.00)$, translation $(t_x, t_y) \sim \mathrm{Uniform}(-0.05, 0.05)^2$ in image-fraction units.
- Horizontal flip with probability 0.5.
- Gaussian blur with radius $r \sim \mathrm{Uniform}(0.5, 1.5)$ px.
- Additive Gaussian noise with $\sigma \sim \mathrm{Uniform}(0, 8)$ on the 0–255 scale.
- JPEG round-trip at quality $q \sim \mathrm{Uniform}\{45, \ldots, 70\}$.

`strong_held_out`

Disjoint operation–parameter-band set: HSV colour rotation, posterisation, sharpening, motion blur, occlusion, and aggressive JPEG compression. JPEG recurs from `strong`, but with a non-overlapping quality band.

- HSV hue rotation by $\Delta h \sim \text{Uniform}(-12°, 12°)$.
- Posterise to $b \sim \text{Uniform}\{4, 5, 6\}$ bits per channel.
- Unsharp-mask sharpen with PIL `ImageFilter.UnsharpMask` (radius 2, percent $\sim \text{Uniform}(120, 200)$, threshold 3).
- Motion blur with a linear kernel of length $k \sim \text{Uniform}\{5, \ldots, 15\}$ px, orientation chosen uniformly from {vertical, horizontal}.
- Random rectangular erasing covering $\alpha \sim \text{Uniform}(0.06, 0.18)$ of the image area, fill colour sampled per call from $\{0, 128, 255\}$.
- JPEG round-trip at quality $q \sim \text{Uniform}\{20, \ldots, 40\}$.

`strong_held_out_2`

Disjoint operation–parameter-band set: gamma, colour-temperature, vignette, salt-and-pepper noise, and high-quality JPEG. JPEG recurs from `strong` and `strong_held_out`, but with a non-overlapping quality band.

- Gamma correction with $\gamma \sim \text{Uniform}(0.6, 1.6)$ (non-linear, distinct from contrast).
- Colour-temperature shift: scale R by $T \sim \text{Uniform}(0.85, 1.15)$ and divide B by T.
- Radial vignette with $\cos^2$ falloff at strength $v \sim \text{Uniform}(0.25, 0.55)$ from image centre.
- Salt-and-pepper impulse noise covering $\rho \sim \text{Uniform}(0.005, 0.025)$ of pixels with value drawn uniformly from $\{0, 255\}$.
- JPEG round-trip at quality $q \sim \text{Uniform}\{80, \ldots, 95\}$ (distinct from `strong`'s 45–70 and `strong_held_out`'s 20–40).

Disjointness. The three families share no operation–parameter-band combination — a property enforced by inspection of the function bodies and re-checked in the code review for every new family proposed. JPEG is the sole recurring transform type; its quality ranges are chosen non-overlapping so that the post-pipeline compression artefact is family-distinguishable. The composite augmentation pipeline is deterministic in the per-pair seed: the same seed reproduces byte-identical perturbed images, which is exploited by the verification harness in `tests/test_public_data_pipeline.py`.

D Bootstrap CI protocol

Every AUROC in the paper is reported with a 1,000-sample percentile bootstrap 95% confidence interval. The implementation is `bootstrap_auroc_ci` in `src/derma_jepa/metrics.py`; the unit tests at `tests/test_metrics.py` cover the AUROC point estimate, the bootstrap CI shape, the EER threshold, and the FPR-at-fixed-TPR computation.

Resampling unit. The resampling unit is the pair: each bootstrap sample draws $N = 2{,}000$ pair indices with replacement from the N stable–changing pairs in a split, recomputes AUROC on the resampled set, and records the value. The 2.5% and 97.5% percentiles of the 1,000 resampled AUROC values define the reported CI. Pair-level resampling matches the unit at which the score s is computed (eq. 5); the resampling is *not* stratified by label, so the per-resample stable/changing balance varies (one standard deviation $\sim\sqrt{N/4} \approx 22$ pairs around the 1,000:1,000 mean). The reported CI is therefore marginal over the empirical class balance; we did not observe class-balance-driven CI inflation in spot checks. Resampling is also not stratified by diagnosis or by anatomical site.

Determinism. The bootstrap RNG is seeded from the same project seed used for split assignment, predictor initialisation, and pair generation. Re-running a configuration with the same seed reproduces byte-identical CIs.

Seed-mean CIs. For the EXP-007 and EXP-008 seed sweeps (§A.10), the reported ± values are unbiased standard deviations across the five seed-mean AUROC estimates; the reported "95% CI[mean]" is the seed-mean point estimate plus and minus 1.96 times the seed-mean standard error ($\sigma/\sqrt{n}$ with $n = 5$). These intervals quantify seed variance and are distinct from the per-pair bootstrap CIs reported on a single seed.

Companion metrics. Alongside AUROC, every run reports AUPRC, the equal-error-rate threshold, and the false-positive rate at a fixed true-positive rate of 0.8. Each is bootstrapped under the same protocol. AUPRC tracks AUROC rank-monotonically across the nine runs; EER and FPR-at-TPR add operating-point information that is sensitive to the train→test score-distribution shift discussed in §7.1. Per-run values are mirrored at `<run_id>/metrics.json` under `abdelstark/derma-jepa-runs`.

E Representation-health checks

Three representation-health diagnostics accompany every run, computed by `_collapse_checks` in `src/derma_jepa/training.py` on the L^2-normalised predictor outputs over the test split. The `collapsed` flag triggers if either the prediction-norm minimum or the dimension-variance minimum falls below 10^{-8} — a representation collapse would render the cosine-distance score ill-defined. None of the nine primary-tier runs trips the flag.

Prediction-norm mean and minimum. The mean and minimum of $\|g_\theta(z(x))\|_2$ across the test-split context inputs $x \in \mathcal{D}_{\text{test}}^{\text{stable}} \cup \mathcal{D}_{\text{test}}^{\text{changing}}$ (un-normalised predictor outputs). A close-to-zero minimum would signal a collapsed direction; a close-to-one mean confirms that L2-normalisation is acting on non-degenerate vectors. All nine runs report mean ≥ 0.97 and minimum ≥ 0.83.

Dimension-variance mean and minimum. $\min_j \operatorname{Var}_{x\in\mathcal{D}_{\text{test}}}(g_\theta(z(x))_j)$ and the analogous mean across the d output dimensions. A near-zero minimum-variance dimension would signal that the predictor has effectively dropped a coordinate. All nine runs report a minimum of order 10^{-3} or larger; the natural-image- backbone runs (EXP-004 / EXP-005 / EXP-006a / EXP-006b) sit one order of magnitude above the DermLIP and BiomedCLIP runs, consistent with the fit-quality gap and not with collapse.

Collapse flag. Binary; true if either of the two minima above falls below 10^{-8}. None of the nine reported runs is flagged. The threshold is conservative: 10^{-8} corresponds to roughly single-precision noise on a unit-norm vector and is intended to catch hard collapse rather than soft anisotropy.

Where the values live. Each run's `metrics.json` → `representation_health` block records the four numerical diagnostics and the binary flag. The aggregator `scripts/aggregate_seed_sweep.py` pulls these fields when computing seed-sweep summaries.

F Compute budget

Every primary-tier run executes on a single NVIDIA A10G 24 GB via Hugging Face Jobs (`a10g-large`). The wall-clock breakdown for the canonical EXP-004 recipe (∼95–100 min end-to-end, measured at ∼98 min from the EXP-004 `progress.jsonl` timestamps) is approximately:

- Manifest build, deterministic-augmentation pair generation, and data-audit checks: ∼10 min.
- Backbone embedding export over 6,000 pairs (×2 sides): ∼10 min for ViT-B/14 and ViT-B/16 backbones.
- Cheap-baseline evaluation (pixel L2, SSIM, raw cosine): ∼5 min.
- Predictor training (200 epochs, batch 128, ∼8 steps per epoch): ∼50 min for the linear predictor; the MLP predictor runs in the same envelope.
- Bootstrap CIs, representation-health checks, report writing, run-archive upload: ∼10 min.

The MLP-on-DINOv2 runs (EXP-005 and EXP-006a) take ∼105 min because the MLP forward/backward is approximately twice the linear predictor's. The DermLIP and BiomedCLIP runs sit in the same ∼95-minute envelope as the linear DINOv2 and CLIP runs: backbone forward dominates only the embedding-export phase, and once embeddings are cached, predictor training is backbone-independent.

Total for the paper. Nine primary-tier runs at ∼95–105 min each plus eight additional seed-sweep runs at ∼95 min each totals approximately 28 A10G-hours of compute, plus an additional ∼3 A10G-hours absorbed by two infrastructure-flake re-launches during the seed-sweep cohort. All accumulated at Hugging Face Jobs free-tier rates; the launchers are committed under `scripts/` and the pinned dependency versions for the hosted environment are at `scripts/hf_jobs_constraints.txt`.

Reproducibility cost. Reproducing the entire nine-run arc plus the seed sweep from a clean checkout requires the same ∼28–31 A10G-hours, plus a one-time HAM10000 download (the source manifest caches it once). No private data, no local GPU.

G Reproducibility checklist

The checklist follows the standard NeurIPS reproducibility template, mapped to specific repository assets.

1. **Code.** Public at https://github.com/AbdelStark/derma-jepa. Pinned dependency manifests at `pyproject.toml` and `uv.lock`; hosted-environment pins at `scripts/hf_jobs_constraints.txt`.
2. **Data.** HAM10000 (Tschandl et al., 2018), accessed from the `abdelstark/ham10000` HF dataset mirror. The HF coordinate is exported as `DERMA_JEPA_HAM10000_REPO` in the launcher scripts under `scripts/`; local manifest construction (lesion-ID-aware splits, pair generation, audit) is implemented in `src/derma_jepa/public_data.py`.
3. **Configurations.** One YAML per primary-tier run under `configs/data/`; every run mirrors its exact `config.yaml` alongside `metrics.json` and `baseline_metrics.json` at <run_id>/ on the run archive (`abdelstark/derma-jepa-runs`).
4. **Seeds and determinism.** Default seed 20260422. Seed controls split assignment, pair generation, augmentation sampling, predictor initialisation, and bootstrap resampling.
5. **Hardware.** Single NVIDIA A10G 24 GB via Hugging Face Jobs (`a10g-large`). Reproduction cost is ∼25 A10G-hours for the nine runs plus the seed sweep (Appendix F).
6. **Hyperparameters.** Listed in Appendix B.
7. **Metrics and uncertainty.** Primary AUROC with 1,000-sample pair-level percentile bootstrap 95% CIs; companion AUPRC, EER, and FPR-at-fixed-TPR. Implementation: `src/derma_jepa/metrics.py`; tests: `tests/test_metrics.py`.

8. **Numbers.** Every quantitative claim in the paper traces to `paper/figures/locked-numbers.json`. The paper Makefile runs `paper/verify_claims.py` before PDF and arXiv-package builds, checking the locked values against the run-archive mirror.
9. **Negative results and ablations.** The full nine-run sequence is reported, including the seven runs whose test AUROC lands in the 0.25–0.33 band across natural-image and general-medical pretraining.
10. **Compute budget.** Reported in Appendix F.
11. **Licence.** Source code under MIT (see `LICENSE`); run-archive artefacts inherit the HF dataset licence.
12. **Intended use.** The methodology and the run archive are research artefacts. The paper makes no clinical or diagnostic claim; the proxy is synthetic and the contamination caveat is explicit (§9).

H Pretraining contamination analysis

This appendix is load-bearing for the paper's central claim. It states the contamination question precisely, lays out the available evidence, and identifies what EXP-007 / EXP-008 can and cannot conclude from that evidence.

H.1 The question

The EXP-007 result — frozen DermLIP plus a linear scaffold reaches test AUROC $= 0.944 \pm 0.003$ on `strong_held_out_2` — is consistent with two distinct hypotheses:

(i) *Dermoscopy-domain transfer.* Pretraining on dermatology image- text pairs (Derm1M) organises the embedding space such that a single linear direction captures stable-pair similarity uniformly across the three nuisance families, including the unseen `strong_held_out_2`.

(ii) *HAM10000 image-level overlap.* Derm1M contains HAM10000 raw images, so DermLIP's embedding space is well-fit to HAM10000 lesion identity in a way that survives our synthetic post-hoc nuisance.

Hypotheses (i) and (ii) are not mutually exclusive: a positive EXP-009 outcome (§8) would support (i) being load-bearing; a low EXP-009 outcome would push the explanation toward (ii).

H.2 Why HAM10000 is almost certainly in Derm1M

The Derm1M data card describes its corpus as "approximately one million publicly-available dermatology image-text pairs aggregated from dermoscopy datasets and dermatology-related web sources" (Yan et al., 2025). The card does not publish a per-source breakdown. HAM10000 (Tschandl et al., 2018) is the largest publicly available dermoscopy dataset by image count; ISIC-archive aggregations (Codella et al., 2019) that virtually all public dermoscopy corpora draw from include the HAM10000 split by default. Excluding HAM10000 from a million-scale dermatology corpus would require an explicit and documented filter; no such filter is documented in the Derm1M description. We therefore proceed under the assumption that DermLIP saw HAM10000 raw images during pretraining.

H.3 What EXP-007 alone can conclude

EXP-007 demonstrates a +0.66 test AUROC swing relative to EXP-006b (OpenAI CLIP) when the only varied axis is the pretraining data. EXP-007 alone cannot distinguish (i) from (ii), and the paper explicitly does not attribute the swing to dermoscopy-domain transfer on the basis of EXP-007 alone.

H.4 What EXP-008 adds and what it does not

EXP-008 introduces a third pretraining-data point: BiomedCLIP's PMC-15M, which is approximately 15× larger than Derm1M but is composed of biomedical figure–caption pairs with a very low fraction of dermoscopy and no documented raw HAM10000/ISIC archive ingestion. The result — test AUROC $= 0.329 \pm 0.012$ — contributes three readings:

- **Scale alone is not the explanation.** Fifteen million biomedical pairs without documented raw HAM10000/ISIC archive ingestion do not unlock the proxy, so the EXP-007 win is not attributable to "more pretraining data," "more medical pretraining data," or any monotonic function of pretraining-corpus size.
- **General medical pretraining is not the explanation.** The +0.04 web → general-medical step is small relative to the +0.62 general-medical → dermoscopy step. The unlock is localised at the dermoscopy-specific corpus boundary.
- **Dermoscopy-vs-HAM10000 remains unresolved.** EXP-008's pretraining corpus differs from EXP-007's in two ways simultaneously — domain narrowness and (likely) HAM10000 inclusion — so EXP-008 cannot separate the two factors. EXP-009 (§8) is designed precisely to break this confound by holding "HAM10000 absent" constant while varying the pretraining-domain narrowness from PMC-15M to dermoscopy.

H.5 Boundary of inference

The strongest claim the paper makes about the EXP-007 swing is the following: *under the protocol of §3, varying only the pretraining-data domain among DermLIP, BiomedCLIP, and OpenAI CLIP — holding architecture, predictor, optimiser, hyperparameters, and data pipeline constant — produces a* +0.66 AUROC *swing localised to the dermoscopy-specific corpus.* The claim is a characterisation of the empirical gradient, not an attribution to domain transfer. The corresponding negative claim — "DermLIP generalises to `strong_held_out_2` because dermoscopy pretraining unlocks nuisance invariance" — is not made anywhere in the paper.

I EXP-009 design (candidate)

This appendix expands the EXP-009 sketch in §8 into the operational detail required to evaluate the proposal. We commit the design — including the decision table mapping outcome bands to claim revisions — in advance so that, should we (or anyone else) choose to run it, the result cannot be re-interpreted post-hoc. We do not commit at this stage to running it ourselves; whether to invest the effort, alter the design, or pursue a different follow-up entirely is a decision we are deferring until after community feedback on this preliminary release.

I.1 Goal

Self-pretrain a vision encoder on a non-HAM10000 dermoscopy corpus, then re-run the EXP-004 recipe on top with the linear predictor. The encoder sees no HAM10000 image during pretraining, so the EXP-007-style positive on `strong_held_out_2` — if it appears — can no longer be attributed to HAM10000 image-level overlap and must reflect dermoscopy-domain transfer.

I.2 Pretraining corpus

Assemble from publicly available dermoscopy / clinical-skin sources, with provenance audit on every component to confirm HAM10000 exclusion:

- ISIC archive components other than HAM10000 (BCN20000 (Combalia et al., 2019), MSK-1 through MSK-5, UDA, Mednode, and similar legacy archives).

- DermNet NZ and DermQuest-derived public images (clinical, not dermoscopic; included to broaden the visual distribution).
- PAD-UFES-20 and 7-point checklist datasets (each cross-checked for HAM10000 overlap by lesion-ID and SHA-256 hash).

Provenance audit: every image must carry a source attribution and a SHA-256 hash. The HAM10000 image-hash set is computed from the source manifest and excluded by hash; any image whose source attribution is ambiguous is excluded by default.

I.3 Pretraining objective

DINOv2 ViT-B/14 architecture (i.e. the same family as the EXP-004 backbone, so any test-AUROC change is attributable to pretraining data, not architecture), trained for $\sim$20 epochs with a JEPA-style or masked-image-modelling objective. Hyperparameters target a $\sim$200,000-step regime on a $\sim$8$\times$ A10G or 1$\times$ A100 budget. Pretrain-quality is checked by linear-probing on a held-out HAM10000 split for diagnosis classification — a sanity check that the encoder has not overfit to its training distribution.

I.4 Probe

After pretraining, the encoder is frozen and the EXP-004 recipe runs on top: linear predictor with identity warm-start, SGD, weight-decay-on-$(W - I)$, 200 epochs, the same lesion-ID splits, the same $1{,}000 + 1{,}000$ pairs per split, the same three-family augmentation pipeline, the same seed. The only varied axis relative to EXP-004 is the pretraining data; the only varied axis relative to EXP-007 is HAM10000 inclusion in that pretraining data.

I.5 Decision table

Identical to Table 6 in §8. Outcome bands and claim revisions:

- Test AUROC ≥ 0.85: dermoscopy-domain transfer is sufficient under this scaffold. Headline survives; contamination caveat downgrades from "unpartitioned" to "ruled out."
- Test AUROC in $[0.50, 0.80]$: partial transfer; both factors contribute. Claim becomes "dermoscopy-domain pretraining helps, HAM10000 image overlap helps more."
- Test AUROC in $[0.30, 0.50]$: HAM10000 overlap drove EXP-007. Claim narrows to "frozen-backbone JEPA unlocks when the backbone has seen the eval dataset, not when it has seen the eval domain."

I.6 Why we share the design before running it

Three concrete costs would gate any launch:

(1) *Provenance audit.* Confirming HAM10000 exclusion on every ISIC component requires a per-image hash check against the HAM10000 manifest — the audit is straightforward but is itself a multi-day engineering task before any GPU is spent.

(2) *Pretrain compute.* DINOv2 ViT-B/14 pretraining on $\sim$200,000 steps requires a non-trivial GPU budget that is not in the free-tier envelope used for the other nine runs.

(3) *Pretrain-quality validation.* A weak pretraining run that never learns dermoscopy structure would produce a low test AUROC that the decision table would mis-attribute to the lower bands when the actual cause is pretrain failure. A held-out diagnostic would be needed to disambiguate.

We share the methodology and the nine-run arc at this stage to invite feedback on the proxy-task design and on whether this partition is the right next step. Whether to ultimately run EXP-009, modify it, or pursue a different follow-up depends on that feedback.